

Data Mining (DS3002)**Final Exam**Date: May 29th 2024

Course Instructor

Miss. Eesha Tur Razia Babar

Total Time (Hrs): 3

Total Marks: 100

Total Questions: 5

Roll No

Section

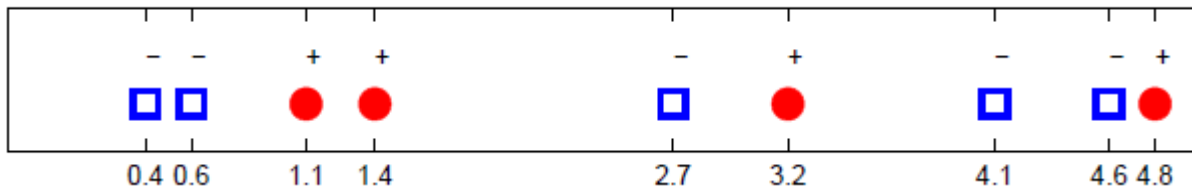
Student Signature

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Attempt all the questions.

Please do not allow the sharing of the calculator.**CLO #1: Understand basic concepts of data mining.****Q1:****Part1: 10 marks**

Given the above data with one scalar feature x (whose values are given below each data point) and a class variable y , with filled circles indicating $y = +1$ and squares $y = -1$ (the sign is also shown above each data point for redundancy), we use a k -nearest neighbor classifier to perform prediction; in the case of ties, we prefer to predict class -1. Answer the following.



(a) Compute the training error rate of a 1-Nearest-Neighbor classifier trained on these data.

It will be zero always because a point's 1 nearest neighbor is that point itself.

(b) Compute the leave-one-out cross-validation error rate of a 1-Nearest-Neighbor classifier on these data.

(c) Compute the training error for a 3-Nearest-Neighbor classifier on these data.

- (b) Compute the leave-one-out cross-validation error rate of a 1-Nearest-Neighbor classifier on these data.

check nearest non-self point.

✓✓ ✓✓ x x ✓ x x

$$\Rightarrow 4/9$$

- (c) Compute the training error for a 3-Nearest-Neighbor classifier on these data.

check nearest 3 points, including self.

✓✓ ✓✓ x x ✓✓ x

$$\Rightarrow 3/9$$

Tick means that point will be predicted right and cross means it will be predicted wrong.

Part2: 10 marks

Suppose that you would like to predict a positive number y from a one-dimensional input feature x . You think that the relationship between y and x is well approximated by an exponential function, $\hat{y}(x) = \exp(ax + b)$. Given m training examples $(x^{(i)}; y^{(i)})$, you want to find regression parameters a ; b that minimize the mean squared error (MSE) in your predictions:

$$J(a, b) = \frac{1}{m} \sum_{i=1}^m (y^{(i)} - \hat{y}(x^{(i)}))^2 = \frac{1}{m} \sum_{i=1}^m (y^{(i)} - \exp(ax^{(i)} + b))^2.$$

- 1) What is the derivative of MSE with respect to a .

- (1) What is $\frac{\partial}{\partial a} J(a, b)$, the derivative of the MSE with respect to parameter a ?

$$\begin{aligned} \frac{\partial}{\partial a} J(a, b) &= \frac{1}{m} \sum_{i=1}^m \frac{\partial}{\partial a} (y^{(i)} - \exp(ax^{(i)} + b))^2 \\ &= \frac{-2}{m} \sum_{i=1}^m (y^{(i)} - \exp(ax^{(i)} + b)) \exp(ax^{(i)} + b) x^{(i)} \end{aligned}$$

- 2) What is the derivative of MSE with respect to b .

(2) What is $\frac{\partial}{\partial b} J(a, b)$, the derivative of the MSE with respect to parameter b ?

$$\begin{aligned}\frac{\partial}{\partial b} J(a, b) &= \frac{1}{m} \sum_{i=1}^m \frac{\partial}{\partial b} (y^{(i)} - \exp(ax^{(i)} + b))^2 \\ &= \frac{-2}{m} \sum_{i=1}^m (y^{(i)} - \exp(ax^{(i)} + b)) \exp(ax^{(i)} + b)\end{aligned}$$

3) Suppose that at some iteration of gradient descent, the regression parameters have values $a = 2$, $b = 5$. Evaluating the derivatives of the MSE at this point, you find that the partial derivative of MSE with respect to a is -2 and the partial derivative of MSE with respect to b is $+3$. After one iteration of gradient descent with step size (learning rate) $= 0.5$, what will be the new values of the a and b parameters?

4) Is the single iteration of gradient descent computed in the previous part guaranteed to reduce the training MSE? Briefly justify your answer.

(3) Suppose that at some iteration of gradient descent, the regression parameters have values $a = 2$, $b = 5$. Evaluating the derivatives of the MSE at this point, you find that $\frac{\partial}{\partial a} J(a, b) = -2$ and $\frac{\partial}{\partial b} J(a, b) = +3$. After one iteration of gradient descent with step size (learning rate) 0.5 , what will be the new values of the a and b parameters?

To minimize, take step in negative gradient:

$$\begin{aligned}a^{\text{new}} &= a - 0.5 \left(\frac{\partial J(a, b)}{\partial a} \right) = 2 - 0.5(-2) = 3 \\ b^{\text{new}} &= b - 0.5 \left(\frac{\partial J(a, b)}{\partial b} \right) = 5 - 0.5(+3) = 3.5\end{aligned}$$

CLO #2: Apply data mining techniques to extract information from large data sets.

Q2:

Part1: 10 marks

Suppose we observe some training data with two binary features x_1 ; x_2 and a binary class y . After learning the model, you are also given some validation data.

Table 1: Training Data

x_1	x_2	y
0	0	0
0	1	0
0	1	1
0	1	1
1	0	1
1	0	1
1	1	0
1	1	0

Table 2: Validation Data

x_1	x_2	y
0	0	1
0	1	0
1	0	1
1	1	0

In the case of any ties, we would prefer to predict class 0.

(a) Give the predictions of a joint Bayes classifier on the validation data. What is the validation error rate?

(b) Give the predictions of a naïve Bayes classifier on the validation data. What is the validation error rate?

(a) Give the predictions of a joint Bayes classifier on the validation data. What is the validation error rate?

$\hat{y}$:

$$00 \Rightarrow 0 \quad \times$$

$$01 \Rightarrow 1 \quad \times$$

$$10 \Rightarrow 1 \quad \checkmark$$

$$11 \Rightarrow 0 \quad \checkmark$$

$$\Rightarrow \text{validation error rate} = 1/2.$$

(b) Give the predictions of a naïve Bayes classifier on the validation data. What is the validation error rate?

$$p(y=0) = p(y=1) = 1/2$$

$$p(x_1=1 | y=0) = 1/2$$

$$p(x_2=1 | y=0) = 3/4$$

$$p(x_1=1 | y=1) = 1/2$$

$$p(x_2=1 | y=1) = 1/2$$

$\Rightarrow$

$$\hat{y}:$$

$$00 \Rightarrow 1$$

$$01 \Rightarrow 0$$

$$10 \Rightarrow 1$$

$$11 \Rightarrow 0$$

$$\Rightarrow \text{validation error rate} = 0.$$

Part 2: 10 marks

Consider the table of measured data given on the right. We will use a decision tree to predict the outcome y using three features, x_1 ; x_2 ; x_3 . In the case of ties, we prefer to use the feature with the smaller index (x_1 over x_2 , etc.) and prefer to predict class - over class +.

x_1	x_2	x_3	y
0	0	0	+
0	0	1	-
0	1	0	-
0	1	1	+
1	0	0	-

- (1) Which attribute has the highest information gain? Justify your answer.
 (2) Based on this choice, build the complete decision tree learned on this data

(1) Which attribute has the highest information gain? Justify your answer. (3 points.)

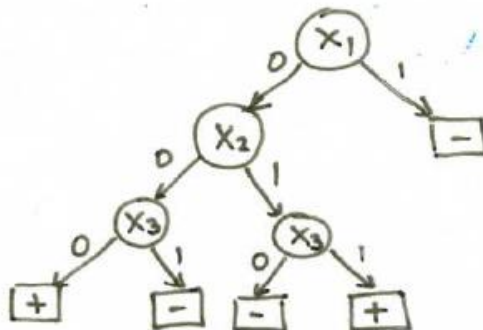
$$H(y|x_1) = \frac{4}{5} H\left(\frac{1}{2}\right) + \frac{1}{5} H(1) = \frac{4}{5} = 0.8$$

$$H(y|x_2) = \frac{3}{5} H\left(\frac{2}{3}\right) + \frac{2}{5} H\left(\frac{1}{2}\right)$$

$$= \frac{3}{5} \left(-\frac{2}{3} \log \frac{2}{3} - \frac{1}{3} \log \frac{1}{3} \right) + \frac{2}{5} = \frac{3}{5} \left(\log 3 - \frac{2}{3} \right) + \frac{2}{5}$$

$$H(y|x_3) = \frac{3}{5} H\left(\frac{2}{3}\right) + \frac{2}{5} H\left(\frac{1}{2}\right) = 0.95 \quad \therefore \boxed{x_1}$$

(2) Based on this choice, build the complete decision tree learned on this data. (3 points.)



CLO #3: Design and implement the efficient and scalable algorithms

Q3:

Part1: 10 marks

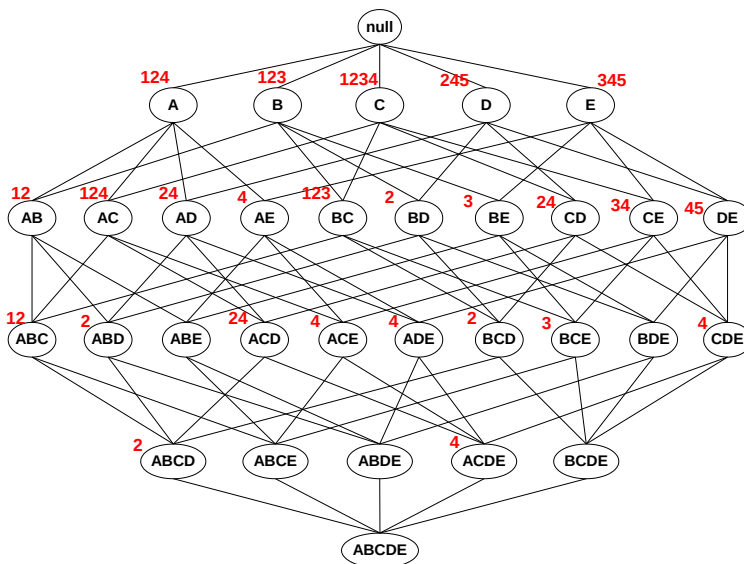
Use the following item sets to construct the FP Tree and then generate frequent item sets through FP tree algorithm using min_support as 2.

TID	Items
1	{A,B}
2	{B,C,D}
3	{A,C,D,E}
4	{A,D,E}
5	{A,B,C}
6	{A,B,C,D}
7	{B,C}
8	{A,B,C}
9	{A,B,D}
10	{B,C,E}

This question is available in slides as it is.

Part2: 10 marks

Use the following figure to solve this question. The number written over each itemset indicates in which transaction that itemset appeared. For example, if 124 is written above itemset A, it means that itemset A appeared in transactions 1,2 and 4.



- 1) Find the maximal item set.
- 2) Find the closed item sets.
- 3) Find item sets that are both closed and not maximal.
- 4) Find item sets that are both closed and maximal.

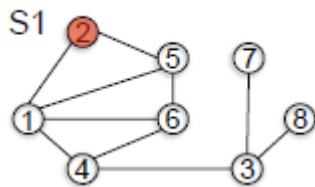
This question is available in slides as it is.

CLO #3: Design and implement the efficient and scalable algorithms

Q4:

Part 1: 10 marks

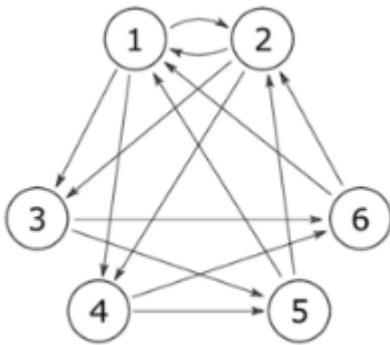
Apply Depth First Search on the following graph. Also, mention both L and M lists at each step.



This question is available in slides as it is.

Part 2: 10 marks

Find the adjacency matrix and graph Laplacian of the following graph.



$$A = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \end{pmatrix} \quad L = \begin{pmatrix} 3 & -1 & -1 & -1 & 0 & 0 \\ -1 & 3 & -1 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 & -1 & -1 \\ 0 & 0 & 0 & 2 & -1 & -1 \\ -1 & -1 & 0 & 0 & 2 & 0 \\ -1 & -1 & 0 & 0 & 0 & 2 \end{pmatrix}$$

CLO #3: Design and implement the efficient and scalable algorithms

Q5:

Part1: 10 marks

A volunteer for the Drug & Alcohol Education Center was investigating the attitudes of CSU students toward binge drinking on campus. A simple random sample of 730 students from all four grade levels was taken. Each student was given the statement “Binge drinking of CSU students has become too frequent and dangerous,” and asked whether they strongly agreed, agreed, had no opinion, disagreed, or strongly disagreed. The following contingency table summarizes the results.

National University of Computer and Emerging Sciences

Lahore - Campus

1) Complete the contingency table.

Grade	Strongly Agree	Agree	No Opinion	Disagree	Strongly Disagree	Totals
Freshman		37	11	92	49	
Sophomore	35	28		59	28	175
Junior	36	32	39	43		190
Senior	48	38	22		14	153
Totals	142		97	225	131	730

- What is the marginal probability of those students who have no opinion?
- Compute the conditional probability that a student disagreed with the statement given they were a junior.
- What is the conditional probability that a student was a sophomore given they strongly agreed?

2) What is the marginal probability of those students who have no opinion?
 $97/730 = 0.133$

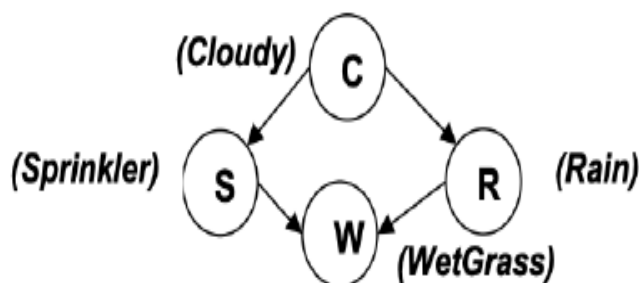
3) What is the interpretation of this marginal probability?
 Approximately 13.3 % of the students had no opinion of the statement.

4) Compute the conditional probability that a student disagreed with the statement given they were a junior.
 $\Pr \{ \text{Disagree} \mid \text{Junior} \} = 43/190 = 0.226$

5) What is the interpretation of this conditional probability?
 Among all CSU juniors, an estimated 22.6% did not agree that binge drinking at CSU had become too frequent and dangerous.

Part2: 5 marks

Below is the Bayesian network of the wet grass problem. Using this network evaluate $P(C = f \wedge R = f \wedge S = t \wedge W = t)$



P(C)
.5

C	P(S)
t	.1
f	.5

C	P(R)
t	.8
f	.2

S	R	P(W)
t	t	.99
t	f	.90
f	t	.90
f	f	.00

$$\begin{aligned} &P(C=f \wedge R=f \wedge S=t \wedge W=t) \\ &= P(W=t \mid R=f \wedge S=t) * P(R=f \mid C=f) * P(S=t \mid C=f) * P(C=f) \\ &= .90 * .8 * .5 * .5 \end{aligned}$$

Part3: 5 marks

True or false: Early stopping can be used to reduce overfitting in neural networks. True

True or false: Increasing the depth of a decision tree classifier will decrease the bias. True

True or false: Increasing k in a k-nearest-neighbor classifier will decrease the bias. False

True or false: Evaluating AUC instead of accuracy will increase overfitting. False

True or false: Training on a single bootstrap of the training data will decrease overfitting. False